

Characterizing miRNA associated with Gastroesophageal Reflux Disease

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SUMMARY

Gastroesophageal Reflux Disease (GERD) is the recurrent reflux of gastric content into the esophagus, resulting in damage to the esophageal mucosa. Twenty percent of adults in the U.S. have GERD and many suffer from further complications such as Barrett's Esophagus and esophageal cancers. Previous studies have shown that many coding RNA (mRNA) have been upregulated in patients with GERD, leading to some of the observed pathologies. The mechanism of upregulation of these mRNA has not been characterized. This study hypothesizes that changes in specific noncoding RNA (miRNA) can regulate the expression of those mRNA. miRNAs are short regulatory RNA that can inhibit translation or promote cleavage of the targeted transcript. Using a publicly available RNA Sequencing dataset, 30 miRNAs were identified that are differentially regulated in patients with GERD. Analysis with various bioinformatic softwares revealed nine miRNAs that had an inverse relationship with the genes associated with GERD (miR4642, miR30D, miR583, miR766, miR149, miR4778, miR4659A, miR4306, miR95). Downregulation of these miRNAs lead to the upregulation of the mRNAs. Continuous upregulation of these mRNAs (CDX2, GDNF, HIF1A, ICAM1, IL1B, IL6, IL8, KRT5, KRT7, NGF, TFF2) have been found to play a role in subsequent diseases associated GERD, including Barrett's Esophagus and esophageal cancers. This study identified miRNAs with robust correlation to genes affected in GERD and subsequent complications. Regulating these miRNAs and their targets could be possible treatments for GERD as well as the aforementioned gastrointestinal diseases.

INTRODUCTION

Gastroesophageal Reflux (GER) is the uprising of gastric content from the stomach through the esophagus, and can cause discomfort in forms such as heartburn [1]. GER is a common occurrence especially under certain conditions such as diets consisting of fatty foods, but it is not persistent. However, a more chronic form of GER known as Gastroesophageal Reflux Disease (GERD) can arise as a result of malfunctioning or damage to the Lower Esophageal Sphincter (LES), among other things such as hiatal hernias and pregnancies. The LES is responsible for maintaining a level of pressure between the esophagus and stomach to ensure that stomach contents remain within the stomach. Factors such as smoking and drinking can damage the LES, as well as obesity which can reduce its pressure, and other malfunctions such as transient LES relaxations (TLESRs) can result in GERD [2-4].

GERD is a prominent disease with about 20% of adults in the U.S. suffering from it [5]. It can also lead to a number of complications with a variety of implications, such as esophagitis and Barrett's Esophagus (BE), which can be fatal and cause ulcers and erosions [6]. Furthermore, GERD has also been found to be associated and correlated with other gastrointestinal diseases/disorders such as celiac disease and Crohn's Disease (if present in esophagus) [7-9].

Treatments for GERD do exist in the forms of temporary relief or surgical procedures which reduce quality of life. The most common medications prescribed are proton pump inhibitors (PPIs) and Histamine 2 (H2) blockers, both of which are responsible for lowering acid production and secretion. The most common surgical procedure for GERD is Transoral Incisionless Fundoplication (TIF), which wraps a portion of the stomach around the esophagus [10]. While the treatment does provide long-lasting relief, it often leads to a number of complications that impede ordinary lifestyles, such as infections, difficulty belching or swallowing, or even tearing of the esophagus [11]. In short, while treatments for GERD do exist, they simply seek to provide temporary relief to physical symptoms, or are surgical procedures with long-lasting negative effects.

A large number of protein coding genes, known as messenger RNAs (mRNA) have been identified to play a functional role in GERD. The proteins encoded by these mRNA consist of mainly pro-inflammatory immune responsive genes, such as cytokines. Several of these proteins have been linked to the Nuclear factor Kappa-Beta (NF- κ B) signaling pathway [12-13].

As previously stated, GERD is caused by inhibition of the function of the LES, whether that be by malfunction, hiatal hernia, or obesity. This inhibition allows for the rising of hydrochloric acid (HCl) from the stomach, making the environment of the esophagus very acidic.

The significant decrease in pH induces the NF- κ B pathway. Proinflammatory cytokines, i.e Tumor Necrosis Factor Alpha (TNF α) and Interleukin-1 Beta (IL-1 β) are responsible for the initial signaling of the NF- κ B pathway [14-16]. Following initial signaling, ion channels such as Acid Sensing Ion Channel 1/3 (ASIC1/3) and Transient Receptor Potential Vanilloid-1 (TRPV1) are promoted to combat the increase in pH and pain sensation from reflux [17-22]. Through a series of protein-protein interactions, the nuclear localization signal of NF- κ B is unmasked, allowing it to translocate to the nucleus and stimulate the production of other proinflammatory proteins, including other interleukins [23-26], Intercellular Adhesion Molecule 1 (ICAM1) [27], *Caudal*-related homeobox transcription factor 2 (CDX2) [28], Cyclooxygenase-2 (COX2) [29], Hypoxia Induced Factor 1 Alpha (HIF1 α) [30-31], Trefoil Factor Family (TFFs) [32], Protease Activated Receptor 2 (PAR2) [18, 33], Nerve Growth Factor (NGF) [18], keratins [16], and a variety of Mucins, all of which have been previously found to be associated with the presence of GERD [34-35].

The mechanisms in which mRNA associated with GERD are regulated in the disease are poorly understood. In this study it is hypothesized that microRNA (miRNA) could regulate mRNA associated with GERD. Future upregulation of these mRNAs via miRNA can result in complications associated with GERD connected with these mRNA.

While mRNA play an important role within the genome, they only comprise a small percentage of its entirety. Around 70% of the genome is transcribed into RNA, with coding RNAs making up merely 1-2% of the genome [36]. The remaining 68-69% is noncoding RNAs, with an example of it being miRNA with an average size of 22 nucleotides for the mature miRNA [37].

MicroRNA (miRNA) regulates mRNA expression at the post-transcriptional level [37]. When the mRNA is synthesized and allocated from the nucleus to the cytoplasm, it can either be used to code for a protein or its expression can be inhibited by miRNA. miRNA is first synthesized as primary -miRNA (Pri-miRNA) by RNA Polymerase II and is folded into secondary structures such as hairpin loop [38]. Nuclease enzymes such as DROSHA and DGCR8 trim the hairpin to form the precursor-miRNA (Pre-miRNA), which is exported to the cytoplasm [39]. Following exportation, the pre-miRNA is further processed in a 22 nucleotide fragment that is incorporated into RNA Induced Silencing Complex (RISC-Loading Complex) [40]. One of the two strands from the duplex is used to target mRNA sequences through complementary base-pairing. If the base-pairing between the miRNA and its targeted mRNA are perfectly complementary to one another, the mRNA is hydrolyzed or cleaved. If the base-pairing between the miRNA and mRNA are not perfectly complementary, then the RISC-Loading

Complex with other factors, will prevent the ribosomes from translocating the mRNA, resulting in translational repression [41-45]. Both possibilities of the miRNA are depicted below (**Figure 1**).

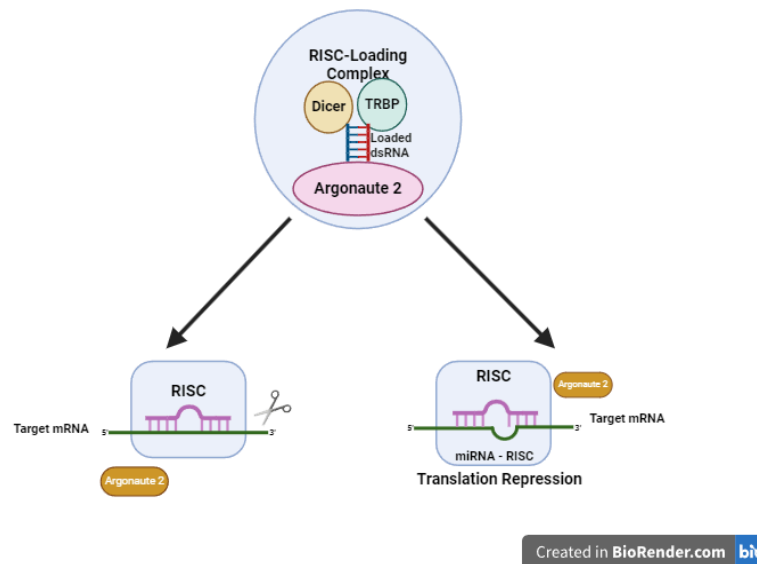


Figure 1: Depiction mRNA degradation via miRNA cleavage or translation repression in the RNAi pathway. Image created in BioRender.com.

This study tests the hypothesis that changes in mRNAs associated with GERD can be attributed to changes in miRNAs that regulate their expression. Since one of the mechanisms by which miRNA regulates its target is through cleavage of the mRNA, an increase in a miRNA should be accompanied with a decrease in the targeted mRNA and vice versa. In this study, the expression between miRNA and targeted mRNA will be correlated to identify an inverse relationship. miRNA-mRNA will then be examined to determine if continued deregulation is associated with other disorders such as Barrett's Esophagus (BE).

RESULTS

To identify miRNA associated with GERD, a publicly available RNA sequencing dataset published by Thomas Greuter, et. al. was used [46]. Among the patients used to isolate total RNA for this dataset were six control patients with no previously identified conditions and six GERD patients only identified to have GERD. Esophageal biopsies were taken from each patient, RNA was isolated from tissues and next generation RNA sequencing was performed.

To further characterize the dataset, Integrated Differential Pathway Analysis (iDEP) was utilized for normalization [47]. The outliers that were too highly or too lowly expressed were

removed. Normalization showed similar distribution of the data between the experimental and control groups (**Figure 2**).

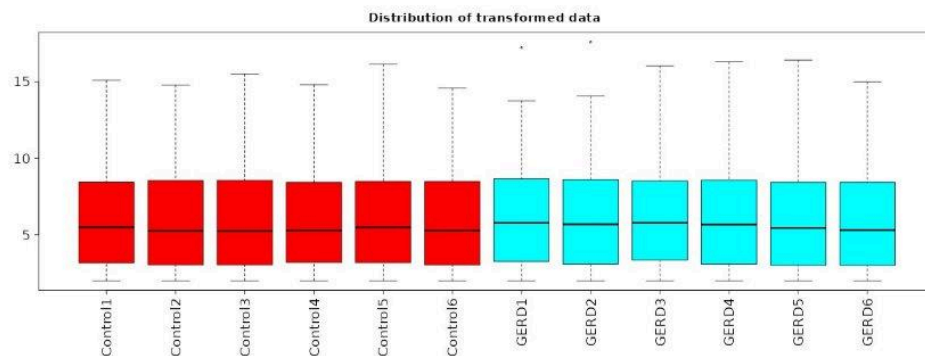


Figure 2: Normal distribution of transformed data from the GSE148381 dataset; retrieved from iDEP.

Additionally, there is a clear separation of the control groups and the experimental groups, as demonstrated by the Principal Component Analysis (PCA) plot, with little variance among patients with GERD, and some outlier groups among the control patients. (**Figure 3**).

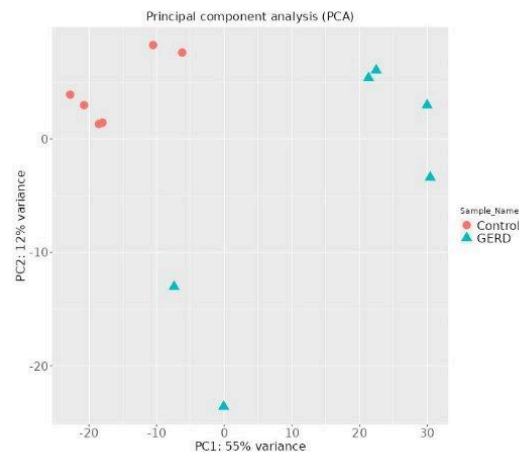


Figure 3: Principal Component Analysis and hierarchical clustering heatmap showing substantial difference in gene expression with little variation among replicates. Red circles indicate control groups, blue circles groups with GERD patients; retrieved from iDEP.

To identify statistically significantly expressed genes, specifically miRNAs, the log2 fold change (Log2FC) was calculated for each gene with TTests done on the Log2 transformed data between the control and experimental data. By doing this, the statistically significant genes were able to be isolated, and their degrees of regulation were also identified. Changes between control and experimental group are further represented with the heatmap, indicating that most

genes that were previously downregulated in control patients were upregulated in GERD patients, and vice versa.

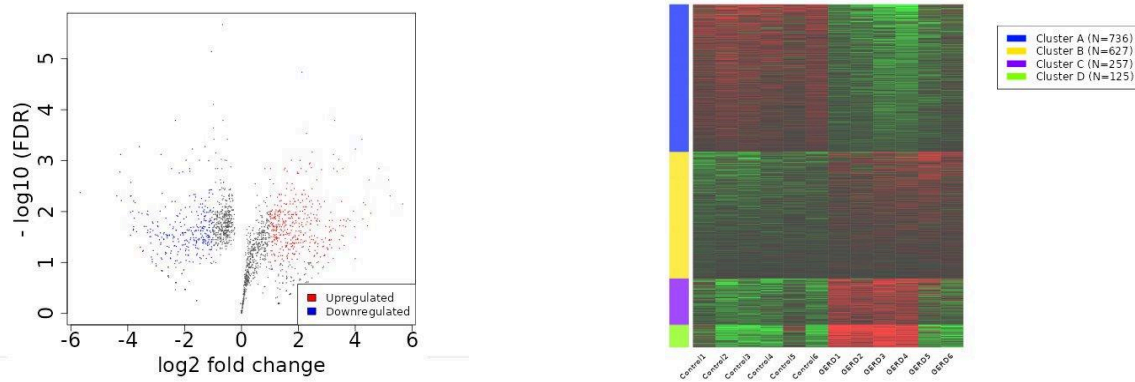


Figure 4: Volcano plot of genes from GSE148381 dataset comparing the degree of regulation to the genes significance (significant if P-Value > .05). Black means insignificant, blue means significantly downregulated genes, and red means significantly upregulated genes.

Further analysis of the dataset revealed the presence of 30 miRNA that were statistically significant, with 18 being downregulated (green) and 12 being upregulated (red) (**Table 1**).

miRNA	LOG2FC	P-Value	miRNA	LOG2FC	P-Value
MIR924	-4.543542573	0.019678823	MIR3619	0.372127263	0.045239018
MIR4642	-3.742738419	0.014084608	MIR22HG	0.456442405	0.045231765
MIR1255B2	-3.635588574	0.049039104	MIR6504	0.757108469	0.049683743
MIR3146	-3.544320516	0.006749696	MIR181A1HG	0.938523052	0.013053529
MIR30D	-2.952866024	0.011831572	MIR581	1.004375526	0.032532964
MIR583	-2.416702468	0.029757932	MIR8063	1.721216397	0.033515985
MIR6508	-2.331205908	0.048248303	MIR505	1.979127024	0.023944529
MIR646	-2.294737062	0.022569562	MIR31HG	2.308625328	0.025431275
MIR4525	-2.085807154	0.0023779	MIR3680-2	2.354052298	0.01908769
MIR766	-2.010745581	0.014633638	MIR4692	2.407288	0.01487991
MIR149	-1.622427235	0.014938513	MIR3664	3.162693432	0.039458295
MIR4778	-1.613415076	0.009793121	MIR4681	3.424983612	0.030216828
MIR4659A	-1.545492596	0.029097162			
MIR1303	-1.337719215	0.016243043			
MIR1299	-1.318450882	0.005981397			
MIR744	-1.013818701	0.014019931			
MIR4306	-0.878880005	0.001211133			
MIR95	-0.767393833	0.038117445			

Table 1: Fold Change and P-Value of all statistically significant (P-Value < .05) miRNA from GSE148381 dataset; Green signifies downregulation while red signifies upregulation.

The list of statistically significant miRNA was scrutinized in order to determine whether they regulated mRNA previously associated with GERD. The expression levels of both were then checked for an inverse correlation, which would signify cleavage of the mRNA. Nine miRNA were identified to be inversely correlated with mRNA associated with GERD (**Table 2**).

The miRNA appears to have a wide degree of regulation, with the miRNA interacting with 5470 nodes in total. (**Figure 5**). This large number of genes that these miRNA interact with further demonstrates the large possibility that the identified miRNA play a role in the pathogenesis of GERD, and the subsequent complications.

miRNA	Expression		mRNA (GERD)	Expression	
	LOG2FC	P-Value		LOG2FC	P-Value
MIR4642	-3.742738419	0.014084608	PAR2, HIF1A	Upregulated	<.05
MIR30D	-2.952866024	0.011831572	IL1β	Upregulated	<.05
MIR583	-2.416702468	0.029757932	IL2	Upregulated	<.05
MIR766	-2.010745581	0.014633638	ASIC1, KRT5, IL5	Upregulated	<.05
MIR149	-1.622427235	0.014938513	ASIC1, CDX2, ICAM1, NGF, IL6	Upregulated	<.05
MIR4778	-1.613415076	0.009793121	GDNF, HIF1A, KRT7, TFF2, KRT5	Upregulated	<.05
MIR4659A	-1.545492596	0.029097162	HIF1A, TFF2	Upregulated	<.05
MIR4306	-0.878880005	0.001211133	IL8	Upregulated	<.05
MIR95	-0.767393833	0.038117445	PAR2, GDNF, ICAM1, IL2	Upregulated	<.05

Table 2: miRNA with fold change are displayed on the left; GERD associated mRNA that they regulate listed on the right. Only inversely correlated mRNAs are listed because inverse correlation shows regulation in GERD. Data retrieved from miRDB.org [48].



Figure 5: Representation of all genes that the identified miRNA could regulate. Blue squares represent miRNAs and purple circles represent nodes/genes that the miRNAs regulate. Image retrieved from miRNET.ca [49].

Common complications of the pathogenesis of GERD include Barrett's Esophagus (BE) Esophageal adenocarcinoma (EAC) and Esophageal Squamous Cell Carcinoma (ESCC). By observing a similar regulation of the mRNA in GERD and its complications, it's possible they play a role in subsequent complications and therefore the miRNA do as well (**Table 3**)

mRNA		Complications	
mRNA in GERD	Expression	Name	Expression
CDX2	Upregulated	EAC	3.70044
GDNF	Upregulated	EAC	0.88
HIF1A	Upregulated	ESCC	0.99
ICAM1	Upregulated	ESCC	1.54
IL1B	Upregulated	BE	2.70044
IL6	Upregulated	BE	4.564785
IL8	Upregulated	BE	3.906891
KRT5	Upregulated	ESCC	9.43
KRT7	Upregulated	EAC	7.0
NGF	Upregulated	ESCC	1.0
TFF2	Upregulated	EAC	5.84

Table 3: Correlation of upregulated mRNA in GERD to their respective degree of regulation in GERD complications. Data retrieved from Genevestigator® and GSE48240 dataset [50-51].

Additionally, if the mRNA are found to work together in proinflammatory pathways, their respective miRNAs could also work together in the pathogenesis from GERD to BE. Therefore, with the use of STRING Functional Protein Association Network [52- 64], a group of mRNAs found to play a role in GERD and subsequent diseases of GERD can be formed. It can be observed that the inflammatory proteins seem to work in multiple pathways with one another in the progression of these diseases. The pathways in which these mRNA interact with one another include signaling pathways in the pathogenesis of BE (**Figure 6**).

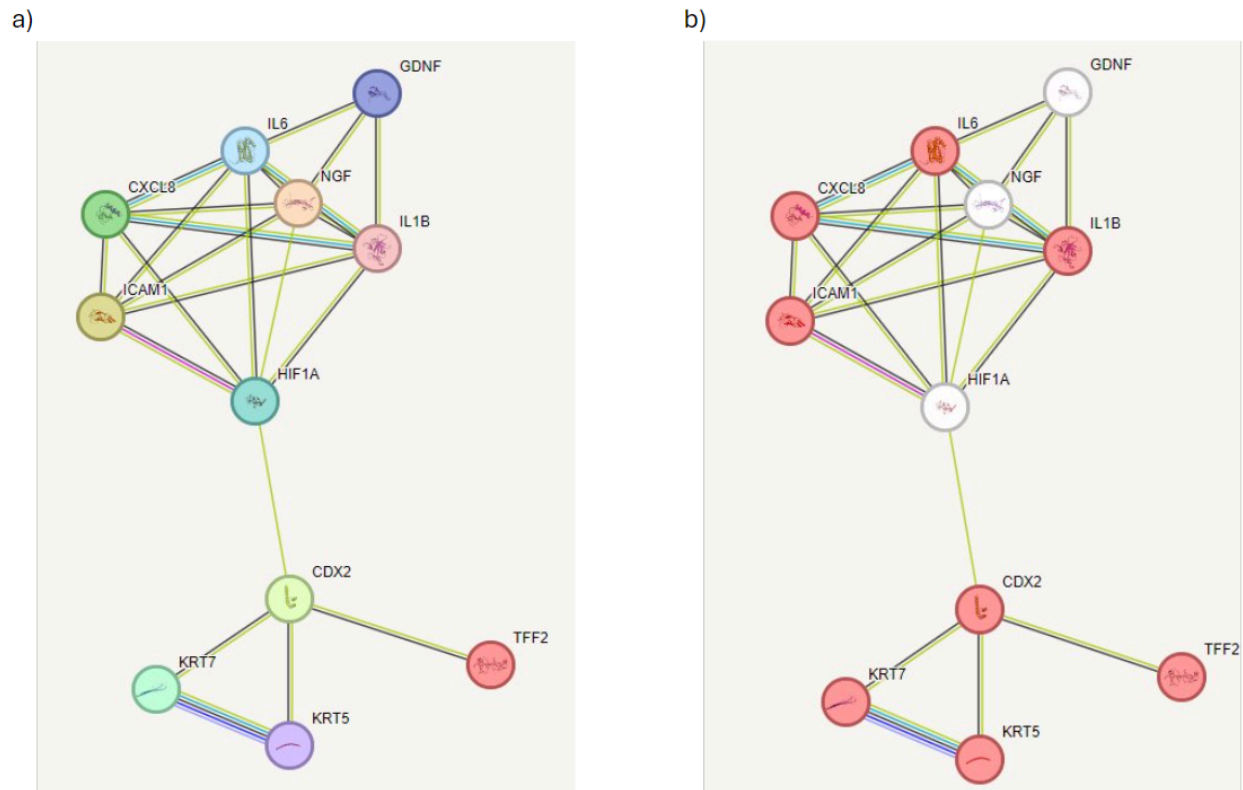


Figure 6: a) Interactions of all mRNA found to be inversely correlated with miRNA and predicted to play a role in complications. b) Specific mRNA found to play a role in BE by a singular study (red). Data retrieved from STRING Functional Protein Association Network.

DISCUSSION

Previous studies concerning the pathogenesis of GERD have identified a wide variety of miRNA that play a role in GERD. Among the miRNAs identified, two are the same as the miRNA identified in this study.

Of the aforementioned studies, the first miRNA is the study by Ajay Bansal, et.al, which identified a number of miRNA to be associated with GERD. Of the miRNA identified, miR149 was determined to be significantly downregulated in the presence of GERD, and its subsequent complications [65]. Previous studies have identified miR149 to be downregulated and serve as a tumor suppressor in gastric cancer, breast cancer, hepatocellular carcinoma, and colorectal cancer [66-67]. Additionally, it has been found to play a role in inflammatory response via the NF- κ B signaling pathway [68]. Because the results from this study concerning the role of miR149 in GERD is supported by other researchers' findings, and it plays a role in a vital

pathway for both GERD and its complications, it is apparent that miR149 likely is an important regulatory factor of the mRNA in GERD, and possibly its complications.

miR30D was also identified to play to be differentially expressed in GERD and its complications by Michael P. Craig, et. al., however, their results conflict with those of this research study. Esophageal biopsies from patients with GERD, BE, and EAC reveal an increase in expression in all of the aforementioned diseases, yet the increase in GERD was not statistically significant [69].

Beyond the discussed studies, other studies examined miRNA associated with GERD and its complications without taking the mRNAs that they are responsible for regulating into consideration. However, this study took such factors into consideration, and if a similar degree of regulation is expressed in the mRNA in subsequent complications of GERD, namely esophageal cancers and BE, then there exists a high likelihood that the identified miRNA possible responsible for regulating the mRNA in GERD are also responsible for regulating such mRNA in manifestations of GERD [70-72].

The mRNA identified to play a role in the NF- κ B pathway, and therefore contribute to GERD seem to do so via the promotion of inflammation as a result of the continued backwash of acidic content into the esophagus. Additionally, data retrieved from Search Tool for the Retrieval of Interacting Genes/Proteins (STRING) Functional Protein Association Network suggests that numerous mRNA play a significant role in the NF- κ B pathway in BE [73]. If the mRNA identified to play a role in GERD have a similar function in the origin and pathogenesis of BE, the miRNA found to possibly regulate the mRNA in GERD would likely have a similar function of regulation in the transformation of GERD to BE.

Further analysis of previous studies reveals that this statement holds true for the mRNA. For example, according to the study by Michael Quante, et. al, IL6 appears to be significantly upregulated in BE, arguing that IL6 is likely responsible for the inflammation that BE typically arises from [74]. Additionally, according to the study by Katerina Dvorakova, et. al, which examined gene expression in biopsies of squamous epithelial cells, IL6 seems to be a likely factor in the pathogenesis of development of BE, and its pathogenesis to further malignancies, namely esophageal cancers. [75]. Similarly, according to Daniel S. Oh, et. al, IL8 is significantly increased as a result of continuous reflux (P -value<.001), and is likely a cause for BE. The deregulation of it could help cure patients with BE or subsequent esophageal malignancies [76]. Furthermore, according to the study by Edyta Korbut, et. al, based on the results from a Real Time Polymerase Chain Reaction (PCR) of the mRNA from subjects with BE, KRT7, TFF2, IL1B, IL2, and IL6 were all found to be upregulated in other gastrointestinal biopsies of patients

with diseases that can lead to BE [23]. ICAM1 was also found to be significantly upregulated in the presence of BE, which, according to Belgin Alasehirli, et. al, means it likely plays an important role in the development and progression of BE [77]. According to Roy W. Phillips, et al., and Gabriel M Groisman, et al., CDX2 appears to be a major biomarker in BE and is likely a protein responsible for the original onset of BE by being triggered from the acidic environment and promoting inflammation [78-80]. HIF1A was also found to be a mediating factor in BE and its complications to esophageal cancers by Qiuyang Zhang, et al. and F. C. Ling, et al., the reason being that HIF1A has been demonstrated to be a pro-inflammatory protein in a wide range of diseases, including BE. These findings suggest that an increase in it could promote these diseases, and a decrease could result in patient relief [81-82]. Upregulation of NGF and GDNF, according to Shu-Chuan Yang, also leads to the onset of BE, specifically via inflammation, as it appears to be immune responsive as a result of the uprising of gastric content from GERD [83-84]. All of these proteins are affected by the NF- κ B pathway. Following activation of IL8 and the promoting of the nuclear localization signal, the NF- κ B molecule is then capable of promoting the production of the proinflammatory cytokines found by other researchers to be upregulated in the presence of BE [14-15, 72]. Therefore, the upregulation of these mRNA as a result of the promotion of the NF- κ B pathway results in the pathogenesis of GERD to its complications.

However, one protein not found to express a similar degree of regulation in subsequent complications of GERD is KRT5. While keratins such as KRT5 have been found to be promoted in the signaling of the NF- κ B pathway, KRT5 itself was not found to be upregulated in BE. On the contrary, according to the aforementioned study by Edyta Korbut, et. al, KRT5 was actually found to be downregulated in patients with BE, and only upregulated in a BE-like environment in mice [23].

In the presence of GERD, the NF- κ B pathway is severely overexpressed due to the acidic environment from the presence of HCL in the esophagus. This development promotes the production of cytokines and other proteins, most of which have been found to be associated with GERD and its complications. The upregulation of these proteins are likely a major cause of the development of manifestations of GERD, evident by the occurrence of extreme inflammation. Since the uncharacterized miRNAs from the GSE148381 dataset inversely correlate with their respective mRNAs in both GERD and its complications, the miRNA likely play a role in the transformation of GERD into its subsequent diseases, namely BE and esophageal cancers.

CONCLUSION

Utilizing the RNA-Seq dataset GSE148381, 30 miRNA were identified to be statistically significant. Of them, nine of them were found to be inversely correlated with a mRNA that they are known to regulate that also play a major role in GERD and GERD pathogenesis. Furthermore, the mRNAs identified by other studies to play a role in GERD seem to express a similar degree of regulation in subsequent complications, which include BE, ESCC, and EAC. Because of this, the correlation between the mRNAs and their respective miRNAs are the same, meaning the miRNAs are likely to regulate the mRNAs in subsequent complications of GERD as well.

MATERIALS AND METHODS

Dataset Search

National Center of Technology Information (NCBI) was scoured under the Geo DataSets tab to identify an RNA Seq dataset concerning GERD. Datasets were scrutinized based on the parameters that there were separate and multiple control and experimental groups, with each control group having persons having no diagnosed conditions, and each experimental group having patients only diagnosed with GERD. Based on these parameters, the GSE148381 dataset by Thomas Greuter, et. al. was identified.

Normalization

The dataset was converted from an XLSX file to a CSV file, then uploaded to Integrated Differential Expression & Pathway analysis (iDEP). Using iDEP, the dataset was normalized, and the counts were ensured to be different between the patients who had GERD and the control patients. iDEP 0.96 is a web-based tool used to analyze RNA-seq data which created the normalization plots, as well as other data visualizations that were used such as PCA plots and heatmaps.

Identifying miRNA

miRNA were isolated from the dataset, and utilizing the formula =TTEST((Control levels) , (GERD patient levels), 2, 2) on Microsoft Excel, the variance of the genes between the control and GERD patients was calculated, and the statistically significant genes (P-Value < .05) were isolated. Additionally, using the formula =LOG((Mean GERD) , (Mean control) , 2), the LOG2FC was calculated, and therefore the degree of regulation of the miRNAs was found. Next, utilizing miRDB.org, the miRNA were inputted to find the list of the possible mRNAs it regulates, and this list was scoured to find any mRNA that have been previously found to be

associated with GERD. The regulations of both the miRNA and mRNA were correlated and an inverse correlation was searched for (to signify cleavage of the mRNA). Additionally, the list of miRNA identified to regulate mRNA were uploaded to miRNET.ca to determine the degree of genes these miRNAs have been identified to regulate.

Identifying mRNA Role in Complications

The list of proteins coded by mRNA that were identified to be regulated by miRNA were uploaded to the Genevestigator® database. Through use of the “Perturbations” tab under “Condition Search Tools,” the complications the identified miRNAs have been found to play a role in, with their degrees of regulation as well (GSE48240 RNA-Seq dataset was utilized as well). Further analysis with STRING: Functional Protein Association Network using the “Analysis” tab revealed specific pathways that multiple genes have been identified to cooperatively play a role in were identified.

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